Mathematics & Statistics | Lancaster University

R Markdown and why I love it

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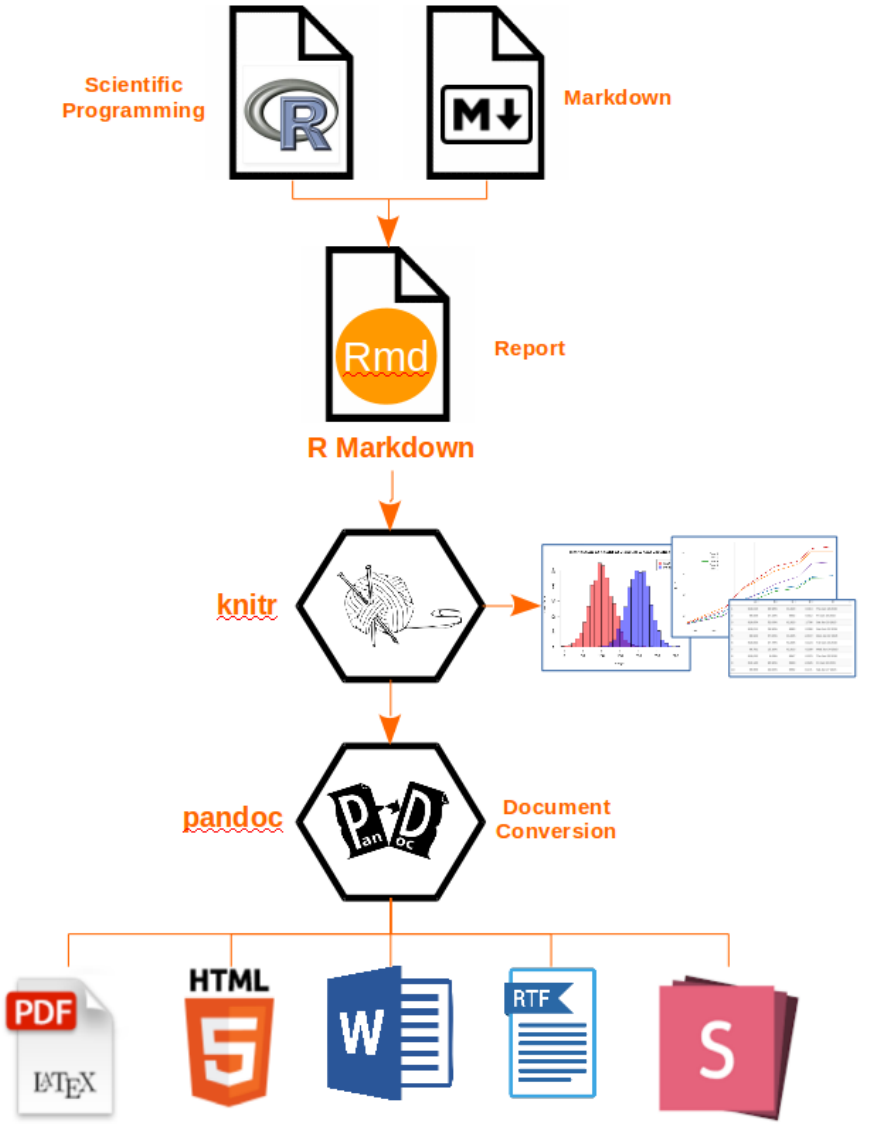
Outline

- What is Rmarkdown?
- Rmarkdown in RStudio
- Rendering to different formats
- Code Chunks
- Short Tutorial
- Advanced features

What is Rmarkdown?

- Plain text file with extension .Rmd
- Reproducible record of your work
- Keeps code and report in a single place

- Documents: HTML, PDF, Word
- Slides: HTML, Beamer Powerpoint
- Other: dashboards, books, websites, shiny documents



Rmarkdown in RStudio

- YAML Header
  - Instructions for how to build the document
- Text
  - Can include parameters from the code
- Code chunks
  - Raw code (R/C/Python) that is run and the output stored/returned
  - three back ticks for starting code chunks, single for inline
    - next to the 1 key!
- Uses markdown syntax but can include direct latex / html if rendering to those formats

Live action Interlude



Short tutorial

1. Create and build the template Rmarkdown document.
2. Change the output to HTML, PDF, Word and view them.
3. Add a code chunk to draw a histogram of the speed variable from the *cars* object - don't include the code.
4. Add some text which displays the mean of the speed variable in the middle of a sentence. Remember single back ticks for inline.

Presentation:

1. Create a presentation (format is your choice)
2. Include the histogram of the speed variable as a whole slide graphic
3. Add the average speed as a bullet point at the top of the histogram.

Advanced Features

- Latex headers
- Hacking slide templates (such as this one)
- Journal article templates https://github.com/rstudio/rticles
- Dashboards and interactive documents (shiny)
- Whole websites and books

My latex lifesaver:

```
includes:
  in_header: header.tex
  keep_tex: true
```

Useful resources

RStudio site: https://rmarkdown.rstudio.com/

Cheatsheet: https://rstudio.com/wp-content/uploads/2015/02/rmarkdown-cheatsheet.pdf

Reference Sheet: https://www.rstudio.com/wp-content/uploads/2015/03/rmarkdown-reference.pdf

Rmarkdown book: https://bookdown.org/yihui/rmarkdown/

Markdown basics: https://rmarkdown.rstudio.com/authoring_basics.html

Hadley's R for data science Markdown chapter: https://r4ds.had.co.nz/r-markdown.html

Graph with bullet

- The mean is 15.4

